

Quantitative assessment of the infection rate of the entomophthoraceous fungus, *Zoophthora anhuiensis* against the green peach aphid *Myzus persicae*.



A two-step method was developed to quantitatively assess the infection rate of the entomophthoraceous fungus, *Zoophthora anhuiensis* (Li) Humber, on the green peach aphid, *Myzus persicae* (Sulzer). Firstly, a standard time-dose-mortality relationship, established by modeling data from bioassay 1 at varying conidial dosages (0.4 - 10.4 conidia/mm²) of *Z. anhuiensis* F97028, was used to yield an estimate of expected mortality probability at a given dosage. Secondly, bioassay 2 was conducted by simultaneously exposing six < or = 4-day-old nymphal colonies to a shower of *Z. anhuiensis* conidia at each of four dosages (resulting from exposures of 0.3 - 8.0 min). Subsequently, the colonies were separately immersed in a 0.1% chlorothalonil solution for 0.5 min to disinfect all surviving conidia on the host integument from 1 - 12 h after exposure under temperature treatments of 15 and 20 degrees C, respectively. The infection rate during a specific period from the end of the exposure to the immersion was then estimated as the ratio of the observed mortality over the expected mortality probability at a particular dosage. The results showed that the infection of *M. persicae* from *Z. anhuiensis* was highly rapid with little difference between aphid colonies maintained at 15 and 20 degrees C before being immersed in the fungicidal solution after exposure. The first 6-hour period after exposure was most crucial to successful infection of the fungus with the infection rate greatly depending on conidial dosages. It took < or = 1 h to infect > 50% of the aphids at a dosage of > 1.5 conidia/mm² and > 90% at > 50 conidia/mm².